

Panel A of Figure 8.9 shows that the cumulated returns (left-hand axis) of the volatility timing and the static 60%/40% strategy. I have scaled the returns of both so that they have a target volatility of 10%. The figure shows that the volatility strategy decisively beat the static strategy and was less prone to drawdowns during the early 2000s and the 2008 financial crisis. During these periods VIX (right-hand axis) was high and the volatility timing strategy shifted into T-bills. It thus partly avoided the low returns occurring when volatility spiked. The cumulated return at the end of the sample is 3.06 for the volatility timing strategy compare to 2.14 for the static 60%/40% strategy. Volatility strategies perform strongly.

An obvious question: if volatility timing does so well, why doesn't everyone do it? Volatility timing requires nimbleness. Panel B of Figure 8.9 shows that the portfolio weights move all over the place. Portfolio weights in equities range from more than 1.5 to nearly zero (as during 2008). Most large investors would have trouble moving in and out of stocks so rapidly or even managing large equities futures positions if done synthetically. So this is not a strategy for all investors.

A deeper question is: what beliefs are implied by the average investor, the representative agent, who holds 100% in the market portfolio and cannot flit in and out of equities by definition? If we rearrange equation (8.9) for the representative agent who holds  $w = 1$  in the market portfolio, then we obtain the capital market line of chapter 6 (see equation (6.1)), repeated here:

$$\mu - r_f = \gamma \sigma^2, \quad (8.10)$$

and high volatilities should correspond to very high future returns (but be accompanied by low contemporaneous returns). We see little predictive power of volatilities in Table 8.7 and overall little evidence of predictability. What the portfolio choice of equation (8.9) is missing, which was used to derive equation (8.10) under the assumption that the representative agent holds 100% of the market, is terms representing liabilities and the long-horizon investment opportunities of chapter 4. Grouping both of these into “hedging terms,” we have

$$w = \frac{1}{\gamma} \frac{\mu - r_f}{\sigma^2} + \text{hedging terms}, \quad (8.11)$$

and the hedging terms must be playing very important roles for the representative agent. Guo and Whitelaw (2006) argue that the hedging terms completely dominate the simple linear risk–return relation and offset the effects of volatility.

A closing comment is that the modern practice of risk management almost exclusively measures risk in (souped-up versions of) GARCH models. I believe risk management is the flipside of expected returns. That is, volatility must be linked with risk premiums, and together they determine valuations. This arises naturally